PROGRAM ASSIGNMENT

Extend a Real Product

Jira Cloud: a dependency and capacity dashboard for SAFe Agile Release Trains, with two teams now and room for six.

Format: Slide deck

Individual: Ramyesh

Tech Cohort 19

01 PICK A REAL PRODUCT

Why Jira Cloud



One ART, six teams, two tools (Mobile runs on Azure DevOps)

1 **Used every day**
We plan sprints and features in it

2 **Big enough to extend**
Boards, plans and a Forge app platform

3 **A real gap**
No view of dependencies and load together

02 THE EXISTING TECH STACK

What Jira runs on, and where our features sit

Screens	React + Atlaskit	Forge (our features)	
Services	Java	Kubernetes	Atlassian Micros
Data	Amazon RDS	DynamoDB	OpenSearch
Speed and events	Memcached	Kinesis	SNS
Cloud and security	AWS	KMS encryption	SSO and JWT

Sources: Atlassian Trust and cloud architecture pages, Atlassian engineering blog, Forge documentation.

03 ADD 1 TO 2 NEW FEATURES

Two features, each one useful on its own

1 ART Dependency Dashboard

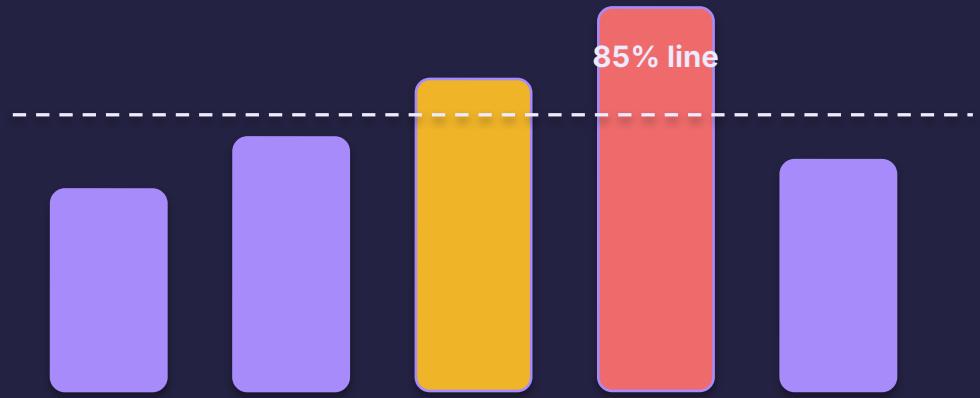
See who waits on whom



Green, amber and red: each dot is one dependency

2 Capacity and Utilisation

Spot overload before the sprint starts



Bars are planned load per sprint, as a share of capacity

04.1 FUNCTIONAL REQUIREMENTS

Feature 1: ART Dependency Dashboard

ID	The system shall...	Priority
FR-1.1	let a team raise a dependency request on another team's item, with a needed-by sprint	Must
FR-1.2	let the receiving team accept (with a sprint) or reject (with a reason)	Must
FR-1.3	show all dependencies for an ART and PI as a team matrix and a sprint timeline	Must
FR-1.4	mark each dependency Green, Amber or Red using the risk rules	Must
FR-1.5	filter by team, status, risk and sprint, and export to CSV	Should
FR-1.6	show Azure DevOps items next to Jira issues, with an ADO badge	Should

Edge cases: a rejected request stays Red until re-planned · circular links are flagged · moved or deleted items become Orphaned · items a user cannot see show as Restricted · outside teams can be added as External.

04.1 FUNCTIONAL REQUIREMENTS

Feature 2: Capacity and Utilisation Planner

ID	The system shall...	Priority
FR-2.1	let a Scrum Master set each member's allocation (0 to 100%) and days off per sprint	Must
FR-2.2	work out team capacity per sprint from its members	Must
FR-2.3	calculate load from planned estimates; utilisation = load ÷ capacity	Must
FR-2.4	show a team-by-sprint heatmap: green 60 to 85%, amber 85 to 100%, red over 100%	Must
FR-2.5	show a person view that adds up load across every team a person works in	Should
FR-2.6	let the RTE test a move of an issue and see utilisation change before saving	Could

Edge cases: allocations over 100% across teams are blocked · zero-capacity sprints show "No capacity" · unestimated issues are counted and flagged · mixed units are never added together.

04.2 NON-FUNCTIONAL REQUIREMENTS

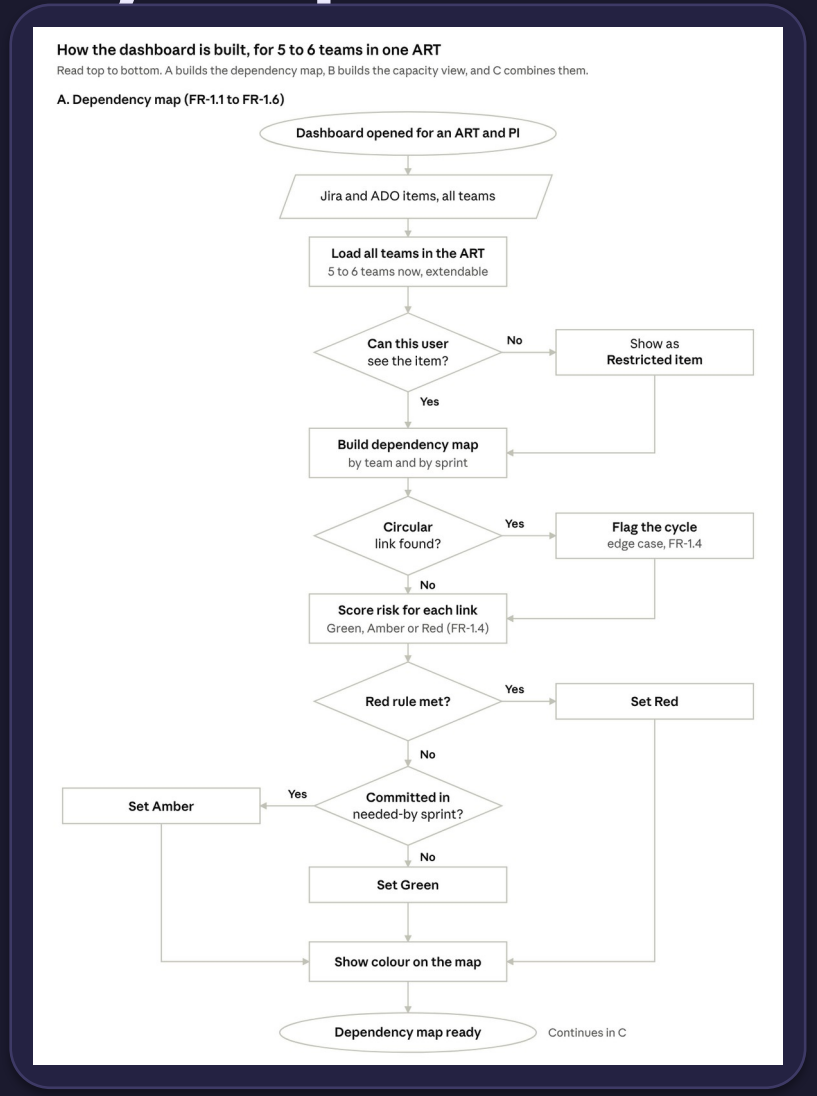
Six quality goals, each with a number

ID	Quality	Target
NFR-1	Performance	Under 3 seconds for 12 teams, 5 sprints and 2,000 issues (95th percentile)
NFR-2	Scalability	Up to 10 ARTs, 150 teams and 20,000 linked issues per site
NFR-3	Availability	99.9% monthly; if the Azure DevOps sync is down, Jira data still shows with a last-synced time
NFR-4	Security and privacy	Every item checked against Jira permissions; data stays in the customer's region
NFR-5	Data freshness	Jira changes within 1 minute; Azure DevOps every 15 minutes
NFR-6	Accessibility	WCAG 2.1 AA; colours always paired with a label or icon

04.3 FLOW DIAGRAM

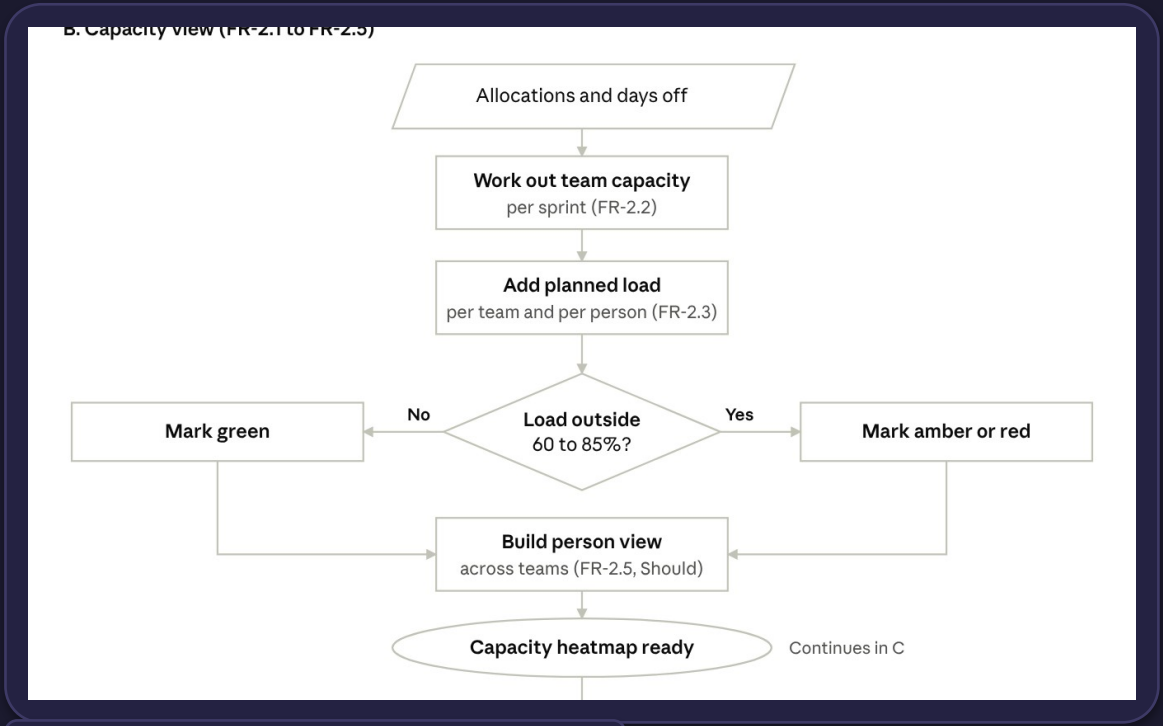
Section A: the dependency map

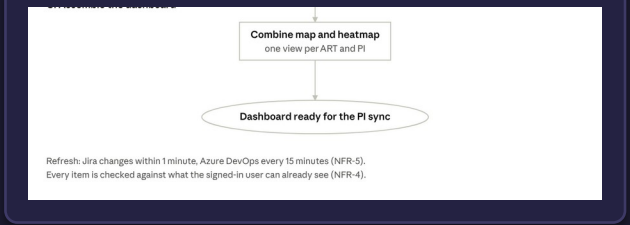
1. Load all teams
2. Check what each user can see
3. Map, then score risk



04.3 FLOW DIAGRAM

Sections B and C: capacity, then one view





1

B Capacity per sprint

Allocations and days off

2

Colour by load

Green 60 to 85%, amber or red outside

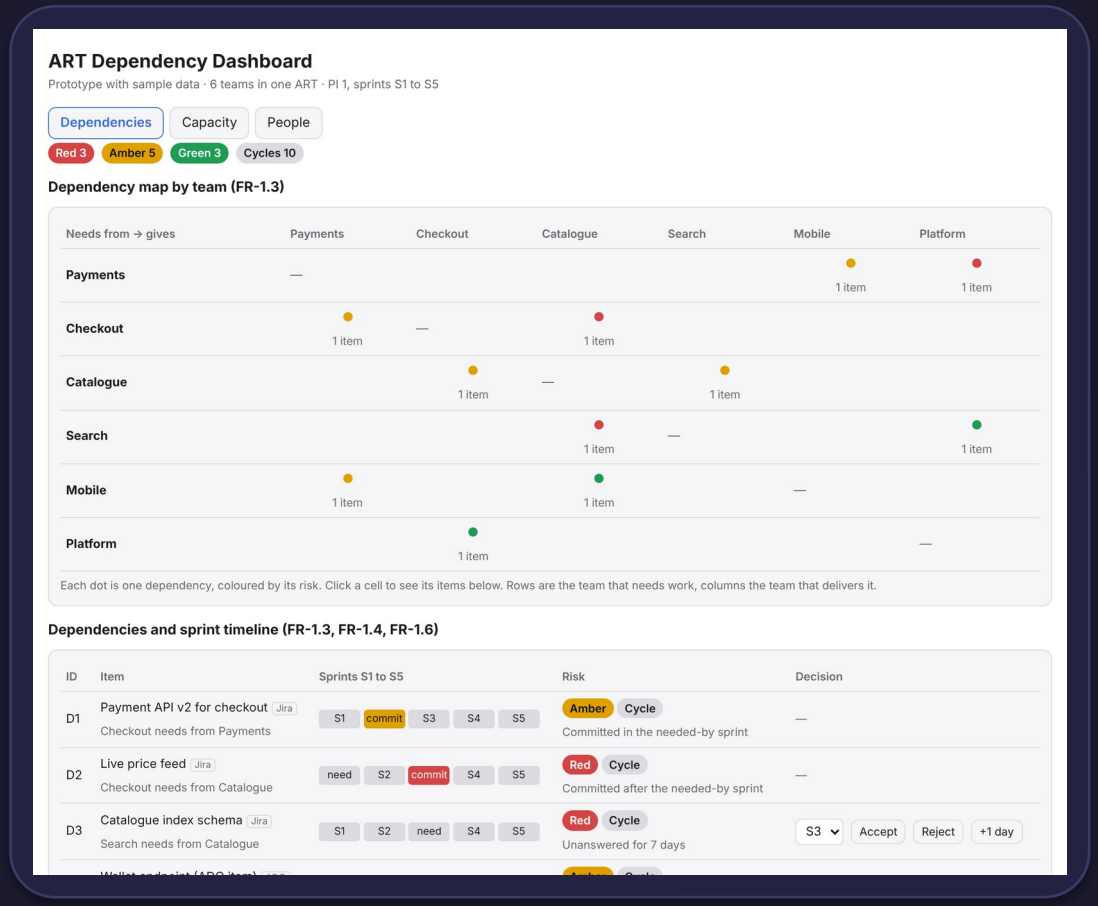
3

C One dashboard

Map and heatmap, refreshed

04.4 WORKING PROTOTYPE

Clickable prototype, sample data



ART Dependency Dashboard

Prototype with sample data · 6 teams in one ART · PI 1, sprints S1 to S5

Dependencies | Capacity | People

Red 3 | Amber 5 | Green 3 | Cycles 10

Dependency map by team (FR-1.3)

Needs from → gives	Payments	Checkout	Catalogue	Search	Mobile	Platform
Payments	—				1 item	1 item
Checkout	1 item	—	1 item			
Catalogue		1 item	—	1 item		
Search			1 item	—		1 item
Mobile	1 item		1 item		—	
Platform		1 item				—

Each dot is one dependency, coloured by its risk. Click a cell to see its items below. Rows are the team that needs work, columns the team that delivers it.

Dependencies and sprint timeline (FR-1.3, FR-1.4, FR-1.6)

ID	Item	Sprints S1 to S5	Risk	Decision
D1	Payment API v2 for checkout Jira Checkout needs from Payments	S1 commit S3 S4 S5	Amber Cycle Committed in the needed-by sprint	—
D2	Live price feed Jira Checkout needs from Catalogue	need S2 commit S4 S5	Red Cycle Committed after the needed-by sprint	—
D3	Catalogue index schema Jira Search needs from Catalogue	S1 S2 need S4 S5	Red Cycle Unanswered for 7 days	S3 Accept Reject +1 day

1 Dependencies

2 Capacity

3 People

claude.ai/artifact/FtacGUD3MnfamMsFHCiaNB

NEXT STEPS

Thank you

1

Real Jira and ADO data

2

A Forge app inside Jira

3

All six teams